Dataset: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Dataset Overview:

The Olist E-commerce dataset originates from a real Brazilian online marketplace and was released publicly to support research into customer satisfaction, delivery logistics, and overall e-commerce performance. It captures detailed information about thousands of customer orders and integrates data from multiple stages of the sales pipeline.

Each order in the dataset is described by:

- Customer information (e.g., location, unique ID)
- Seller information (e.g., location, seller ID)
- Product details (e.g., category, price, dimensions)
- Order timelines (e.g., purchase date, shipping deadline, delivery date)
- Payment records (e.g., type, number of installments, value)
- Customer review data (e.g., scores, comments, review timestamps)

In total, the combined dataset contains over 119,000 rows and 40 columns, drawing from multiple original sources such as orders, payments, reviews, products, customers, and sellers. It provides a rich and realistic foundation for a wide range of e-commerce analytics tasks.

Due to the integrated nature of the data, this dataset allows us to trace the complete lifecycle of an online order—from purchase to delivery to customer feedback. For the purposes of our capstone project, we are leveraging this combined dataset to explore trends, identify pain points in the buyer-seller journey, and generate actionable insights for improving platform performance and customer experience.

Each row in the dataset represents an item within an order and can be linked to other entities (customer, seller, product) through shared identifiers. The dataset also includes English translations for product category names, allowing for easier analysis and visualization.

Research Questions:

1. What key factors lead to late deliveries, and how can we proactively identify high-risk orders to improve customer satisfaction?
2. Which sellers consistently delay order handoffs to carriers, and what underlying patterns or behaviors contribute to this? Can early seller delays help us anticipate customer-facing delivery issues?
3. How do customer segments—such as first-time vs. repeat buyers or by region—differ in cancellation and return behavior, and can we predict these actions early based on browsing or shopping patterns to reduce churn?